

Cross-Asset Statistical Arbitrage Framework

Interview Preparation Dossier

Commodity Futures · Sector ETFs · Multi-Alpha Signals · DCC-GJR GARCH
Turnover-Penalized MV Optimizer · Ledoit-Wolf Covariance · Book/Allocator Architecture

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SECTION 1 — RESUME BULLET BREAKDOWN

"Developed and evaluated multi-alpha signals across commodity futures and sector ETFs, including short-term reversal, medium-term momentum, and cointegration-based spreads (rolling ADF with ECM half-life filtering); aligned signals to expected returns via Fama–MacBeth regressions, and combined them using a turnover-penalized mean–variance optimizer with Ledoit–Wolf covariance estimation and regime conditioning via DCC-GJR GARCH."

Multi-alpha signals across commodity futures and sector ETFs

13 liquid instruments: energy/agricultural/metal futures (CL, NG, Gold, Copper, Corn, Soybeans) and US sector ETFs (XLE, XLF, XLV, XLI, XLB, XLK, SPY). The cross-asset combination is intentional: the two groups have different vol regimes, different carry dynamics, and genuinely different economic drivers — which is exactly what you want for mean reversion. Without genuine heterogeneity, spread signals are just correlated noise.

***Say it as:** "The universe spans commodity futures and US sector ETFs. The cross-asset mix creates genuine economic heterogeneity — oil and energy ETFs are linked but driven by different mechanisms: one is a futures contract with roll dynamics, the other is an equity claim on firms that produce oil. That difference is what creates mean-reverting spreads worth trading."*

Short-term reversal (5-day window)

Signal = $-\text{rolling_sum}(\text{residual_ret}, 5d)$. Beaten-down assets receive high signal → expected reversion. This is a liquidity-provision signal: buy what the market has temporarily oversold. The 5-day window avoids intraday bid-ask bounce while being short enough not to contaminate the 60-day momentum signal. Applied in a dedicated short-term Book with daily rebalancing and lower inertia ($\kappa=2$).

***Say it as:** "Reversal is essentially a liquidity-provision bet. When a sector has been sold off over 5 days relative to its market exposure, there's a mean-reversion expectation. We keep it in a separate daily-rebalancing book because it can't share an optimizer with the weekly momentum signal."*

Medium-term momentum (60-day, 5-day skip)

Signal = $\text{rolling_sum}(\text{residual_ret}, \text{from } t-65 \text{ to } t-5)$. The 5-day skip prevents the short-term reversal effect from contaminating the momentum signal at short lags. Captures persistent trends from behavioral under-reaction, institutional flows, or fundamental momentum in commodities. Cross-sectionally normalized so the signal ranks all 13 assets by trend strength.

***Say it as:** "The 5-day skip is important — without it, you capture the reversal at short lags, which works against the momentum signal. With the skip, you're measuring clean medium-term trend."*

Cointegration-based spreads: rolling ADF with ECM half-life filtering

For each (ETF, commodity) pair: (1) rolling OLS estimates a time-varying hedge ratio in a 120-day window; (2) rolling ADF tests stationarity within that window ($p < 0.10$); (3) ECM regression estimates p — the mean-reversion coefficient from $\Delta \text{spread} = \alpha + p \cdot \text{spread_lag} + \varepsilon$; (4) half-life filter $h = -\ln(2)/\ln(1+p)$ must be < 30 days. All three gates must pass for a signal to be generated that day. When any gate fails, the signal is zero — the pair is not tradable that period.

Say it as: "The three gates eliminate spurious cointegration. ADF alone over-rejects the null in small cross-sections. The ECM filter ensures ρ is in $(-1,0)$, meaning the spread genuinely reverts. The half-life filter removes pairs that are technically stationary but revert over 6 months — too slow for a weekly holding period."

Aligned signals to expected returns via Fama–MacBeth regressions

CS-normalized signals are dimensionless ($\text{std} \approx 1$). Weekly returns have $\text{vol} \approx 0.02$. Without scaling, the optimizer's risk term (in return^2 units) is $\sim 50,000\times$ the alpha term, producing near-zero positions. FM fix: at each date t , run 52 cross-sectional OLS regressions (one per past week) — $\text{forward_return}(t) = \alpha + \beta(t) \cdot \text{signal}(t)$. FM $\beta = \text{mean}(\beta(t-52, \dots, t-1))$, clamped at 0. Scaled alpha = FM_beta $\times$ signal_t, now in expected-return units compatible with the covariance matrix.

Say it as: "The FM step is non-negotiable once you have a covariance-based optimizer. Without it, the optimizer sees signals 14,000 times smaller than the risk terms and produces trivially zero positions. The beta floor at zero prevents the small-cross-section estimation noise from inverting the signal."

Turnover-penalized mean–variance optimizer

Objective: $\max \alpha'x - (\gamma/2)x'\Sigma x - (\kappa/2)\|x - x_{\text{prev}}\|^2 - \lambda \cdot \|\text{sign}(x_{\text{prev}})\|_1$. Parameters: $\gamma=20$ (risk aversion), $\kappa=10$ (position inertia), $\lambda=0.0002$ (transaction cost). Closed-form: $x = (\gamma\Sigma + \kappa I)^{-1}[\alpha + \kappa x_{\text{prev}} - \lambda \cdot \text{sign}(x_{\text{prev}})]$. Post-solve: dollar neutrality ($x := x - \text{mean}(x)$) and position clipping ($|x| \leq 0.07$).

Say it as: "The κ term is what makes the optimizer realistic. Without it, you turn over 100% each week chasing noisy signals. With $\kappa=10$, the optimizer resists position changes unless the alpha strongly justifies them. It implicitly penalizes the quadratic trading cost of getting into and out of large positions rapidly."

Ledoit–Wolf covariance estimation

Applied to residual (beta-neutralized) returns on a 120-day rolling window. Shrinkage toward $(\mu/N) \cdot I$ with analytically optimal shrinkage intensity. $T/N \approx 9.2 \rightarrow$ aggressive shrinkage, which systematically underestimates portfolio vol. Role in the system: controls relative asset sizing (which assets to overweight), not portfolio-level leverage. Absolute leverage is controlled separately by EWMA realized-vol scaling. Critical: must use residual returns, not raw returns — using raw returns inflated Σ by $30\times$ and broke vol targeting in the original code.

Say it as: "LW controls the relative weights within the portfolio — which assets to emphasize given their covariance structure. It's NOT doing our vol targeting. That's handled separately by EWMA scaling on realized PnL, because LW with 120/13 ratio is too aggressive for model-based vol estimation."

Regime conditioning via DCC-GJR GARCH

Pipeline: (1) GJR-GARCH per asset (captures leverage effect: negative shocks increase vol more); (2) DCC on standardized residuals $\rightarrow$ time-varying R_t ; (3) extract ρ_t (mean off-diagonal) and disp_t (std off-diagonal); (4) rolling z-scores vs 60-day baseline; (5) map to 5 categorical labels: normal / clustered / crowded / crisis / broken. Each regime maps to per-book alpha multipliers that control strategy exposure pre-optimizer.

Say it as: "DCC captures that correlations spike during crises — all assets move together, spreads break down. The five-state design distinguishes between elevated-but-dispersed correlations (sector rotation, where spreads might actually widen more) and elevated-and-compressed correlations (crowding, where stat arb is dangerous). That distinction is economically important."

SECTION 2 — "TELL ME ABOUT THE PROJECT" (2-MINUTE ANSWER)

Use this answer verbatim or adapt it. Aim for a calm, conversational pace — technically credible but not rehearsed. Cover original → problems discovered → architectural evolution.

"So the project started as a fairly standard cross-asset stat arb framework. The universe is commodity futures and US sector ETFs — about 13 instruments. The core idea is that these assets share genuine economic linkage — oil prices affect energy ETFs, metals affect industrials — which creates tradable mean-reverting spreads. But the heterogeneity is real: a commodity futures contract has roll dynamics and supply-demand risk that an ETF doesn't, so the diversification is genuine.

The original setup had three alpha signals: short-term reversal (a liquidity-provision bet on 5-day oversold assets), medium-term momentum (60-day residual trend with a 5-day skip to avoid reversal contamination), and cointegration-based spreads using rolling ADF tests and an ECM half-life filter. All three signals were beta-neutralized against SPX first, so we're working on residual returns. They fed into a turnover-penalized mean-variance optimizer with Ledoit-Wolf covariance, and the regime overlay used DCC-GARCH to detect correlation regime shifts.

Now — the more interesting story is what we found when we audited the original implementation seriously. There were three significant issues. First, the carry signal was incorrectly implemented — it used rolling residual returns as a carry proxy, but carry for futures means term-structure roll yield, which we didn't have data for. That signal was pure noise and got removed entirely.

Second, the cointegration test was run on the full sample — a classic look-ahead bias. At any historical date, the model was using future data to decide whether the spread was stationary today. Every spread signal was contaminated. That was replaced with strictly rolling ADF.

Third — and this was the most architecturally important discovery — the covariance matrix was estimated on raw returns, but our positions were beta-neutral. We'd removed the market-beta component from the returns before building positions, but the covariance model was still describing the market-beta risk we no longer had. That caused vol-targeting to be off by a factor of 30. The model thought the portfolio had 0.86% annual vol; it actually had 27.8%. The vol-targeting scalar was trying to leverage up by 30× and exploded into the position limits.

Fixing the covariance mismatch improved Sharpe from 0.07 to 0.25. But it also revealed that a small cross-section of 13 assets with 120 days of history means Ledoit-Wolf shrinks very aggressively toward identity — so model-based vol targeting is fundamentally unreliable here. The solution was to separate the problem: use LW only for relative asset sizing, and do portfolio-level vol control separately via EWMA realized-vol scaling.

The final architectural evolution was the multi-book redesign. Reversal can't share an optimizer with momentum — they operate on completely different horizons. A 5-day signal needs daily rebalancing with low inertia; a 60-day momentum signal is fine with weekly rebalancing and high inertia. Combining them in one optimizer produces turnover distortion in both directions. The solution was a Book abstraction — each strategy owns its own optimizer, covariance, and vol target — orchestrated by an Allocator that handles regime-driven exposure control at the strategy level."

SECTION 3 — PROJECT EVOLUTION

Original Architecture (Archive)

- Monolithic pipeline: `alpha_construction.py` (609 lines) + `optimizer.py` (389 lines).
- Four signals: Reversal, Momentum, Carry, Cointegration — all combined statically.
- Carry implemented as 30-day rolling residual return (incorrect — no term-structure data).
- ADF test run on FULL sample — look-ahead bias at every backtest date.
- Risk-parity weights estimated on full-sample vol — another look-ahead bias.
- LW covariance on RAW returns, vol-targeting scale factors 5–30×, positions explode.
- No separation between alpha logic and portfolio logic — untestable, unmaintainable.

Phase 1 — Carry Removal + Cointegration Fix

- Decision: remove carry entirely. Do not attempt to fix without term-structure data.
- Rolling ADF (120-day window) replaces full-sample ADF. Strictly backward-looking.
- ECM half-life gate added: $\rho \in (-1, 0)$, $h < 30$ days.
- Rolling z-score normalization within each window (not full-sample).
- PnL computation fixed: was capturing only 1 day per week, now accumulates all 5.
Result: Phase 1 Sharpe: 0.07 (still broken — vol 3.7× above target). Source identified: covariance mismatch.

Phase 2a — FM Alignment + Rolling Sharpe Combination

- FM alignment converts dimensionless z-scores to expected-return scale. Solves 14,000× mismatch.
- Rolling 52-week Sharpe weighting ($v=0.3$ shrinkage, 4-week EWM smoothing) replaces static risk-parity.
- DISCOVERY: Reversal has horizon mismatch with weekly optimizer ($\kappa=10$). Excluded from combined alpha.
- Reversal kept as standalone diagnostic for separate short-term book development.

Phase 2b — DCC-GJR GARCH Regime Detection

- GJR-GARCH per asset → standardized residuals Z → DCC → time-varying R_t .
- 5 categorical regime labels: normal / clustered / crowded / crisis / broken.
- First attempt: scalar alpha multiplier POST-solve. This was WRONG — vol targeting re-inflates positions, canceling regime adjustment.
- Design decision: regime control must be PRE-optimizer (alpha multipliers, not position scalars).

Phase 2c — Covariance Root Cause Fix + EWMA Vol Targeting

- ROOT CAUSE: LW on raw returns includes market-beta variance. Positions are beta-neutral — no market exposure.
- Consequence: model vol $\approx 0.86\%$ vs realized 27.8% — 30× mismatch. Vol targeting scalar explodes.
- Fix: Use `residual_ret` (beta-neutral) for LW. But $T/N=9.2$ → aggressive shrinkage → still underestimates.
- Solution: Separate vol control into two layers. LW = relative sizing. EWMA realized-vol = absolute leverage.
- EWMA power-scaling: $\text{scale} = \text{clip}((\text{target_vol}/RV)^2, \text{SCALE_MIN}, \text{SCALE_MAX})$. Power=2 for faster response.

- Cap-aware bounding: $\text{scale_cap} = \text{MAX_WEIGHT} / \max(|x|)$ prevents position limit violations post-scaling.
***Result:** Phase 2c Sharpe: 0.254 (+260% vs Phase 1). Max DD: -7.2% (from -48.8%). Vol: 27.8% (still above 15% target).*

Phase 3 — Modular Engine Architecture

- 937-line monolith split into engine/ (pure functions) + strategies/ (entry point).
- alphas/: pure functions (build_momentum, build_reversal, build_spread).
- portfolio/: optimizer.py (chernov_weights), Book class, Allocator class.
- regime/: compute_regime_signals(), regime_mapping.py (pure function for book decisions).
- Book abstraction: owns alpha, covariance, optimizer params, vol target. Independently backtestable.
- Allocator: orchestrates Books, sums PnL. Regime integration designed, not yet wired up.
- run_baseline.py frozen as regression reference. run_engine.py uses modular components.

Phase 3.5 — Short-Term Book Fix

- ISSUE: Short-term reversal book used log returns. Commodity moves of $\pm 50\%$ (NGc1, 2022) $\rightarrow \log(1.5)=0.405$ vs simple 0.5 $\rightarrow 19\%$ distortion $\rightarrow$ PnL sign flip.
- FIX: Short-term book uses simple returns. Medium-term unaffected (aggregates 5 days per week, dampens extremes).
- Regime gating replaced: hardcoded conditional $\rightarrow$ official get_book_actions() policy (all regimes active with multipliers).

***Result:** Current state: 2 active books (medium_term weekly, short_term daily). Regime framework built, integration pending.*

SECTION 4 — DEEP PROJECT UNDERSTANDING

Economic Thesis

Commodity futures and sector ETFs share fundamental economic linkages (oil→XLE, metals→XLB) that create structural co-movements with occasional mean-reverting deviations. The thesis: these deviations revert on a 1–4 week horizon and can be captured net of transaction costs with disciplined risk management. The cross-asset design creates natural hedging: equity ETF beta is neutralized via SPX regression, leaving a residual driven by sector/commodity-specific factors — the part stat arb can exploit.

Beta Neutralization

Rolling OLS over 60 days: $r_{\text{asset}} = \alpha + \beta \cdot r_{\text{SPX}} + \varepsilon$. Residual ε is the beta-neutralized return used for all signal construction and covariance estimation. Beta is time-varying (60-day rolling) to adapt to shifting factor exposures. Without beta neutralization, momentum and reversal signals would be dominated by market-wide moves rather than idiosyncratic sector/commodity dynamics.

Rolling ADF + ECM + Half-Life (Three Gates)

Gate 1: ADF test within 120-day window at $p < 0.10$ (10% significance). Gate 2: ECM regression — $\Delta \text{spread} = \alpha + \rho \cdot \text{spread_lag} + \varepsilon$ — requires $\rho \in (-1, 0)$. Gate 3: half-life $h = -\ln(2)/\ln(1+\rho) < 30$ days. Each gate eliminates a different failure mode: spurious stationarity (ADF), explosive behavior (ρ gate), and slow reversion (half-life gate). Signal = rolling z-score of spread deviation within window.

Fama-MacBeth Alignment

At each Friday t , run 52 cross-sectional OLS regressions on past weekly data: $r_{\text{forward}}(\tau) = a + \beta(\tau) \cdot \text{signal}(\tau)$ across 13 assets. FM beta = mean(β values), clamped ≥ 0 . This beta $\approx \text{IC} \times \sigma_{\text{returns}}$ — it converts the information coefficient into an expected-return scale factor. Scaled alpha = FM_beta $\times$ signal_t has units of weekly expected return, compatible with the LW covariance matrix. Beta floor = 0 prevents estimation noise in a small cross-section from inverting signal direction.

Optimizer: Objective + Parameters

$\max \alpha'x - (\gamma/2)x'\Sigma x - (\kappa/2)\|x - x_{\text{prev}}\|^2 - \lambda \cdot \|\text{sign}(x_{\text{prev}})\|_1$. Closed-form: $x = (\gamma\Sigma + \kappa I)^{-1}[\alpha + \kappa x_{\text{prev}} - \lambda \cdot \text{sign}(x_{\text{prev}})]$. $\gamma=20$: risk aversion (controls position size). $\kappa=10$: inertia (penalizes turnover, creates position persistence). $\lambda=0.0002$: L1 transaction cost (discourages closing existing positions). Post-solve: dollar neutrality $x := x - \text{mean}(x)$, clipping $|x| \leq 0.07$. $\kappa/\gamma = 0.5$ means the optimizer balances risk-reduction and turnover-reduction approximately equally.

Ledoit-Wolf Covariance

$\Sigma^{\text{LW}} = (1-\alpha)S + \alpha(\mu/N)I$ where α is analytically optimal shrinkage intensity. With $T/N = 120/13 \approx 9.2$, shrinkage is aggressive — small eigenvalues are pulled up significantly. Role: controls RELATIVE asset sizing within the portfolio (which assets to hold more). Role: does NOT control portfolio-level leverage (that's EWMA realized-vol). Must use residual returns (beta-neutral): using raw returns inflated Σ by 30 $\times$, breaking vol targeting.

Vol Targeting: Two Layers

Layer 1 (LW): Relative asset sizing via cross-asset covariance. Layer 2 (EWMA): Portfolio-level leverage. scale = clip((target_vol/RV) 2 , 0.2, 1.5). RV = EWMA of past weekly PnL 2 (halflife=2 weeks). Power=2 for faster response

to vol spikes (when $RV=1.5 \times \text{target}$: power=1 $\rightarrow$ scale=0.67; power=2 $\rightarrow$ scale=0.44 — much more aggressive deleveraging). Cap-aware bounding: $\text{scale_cap} = \text{MAX_WEIGHT}/\max(|x|)$ prevents position limits being violated post-scaling.

GJR-GARCH + DCC

GJR-GARCH: $\sigma^2_t = \omega + (\alpha + \gamma I_{t-1})r^2_{t-1} + \beta\sigma^2_{t-1}$. Asymmetric: $I=1$ when $r<0$ (leverage effect). Fit per asset $\rightarrow$ standardized residuals Z . DCC: $Q_t = (1-a-b)Q_{t-1} + a \cdot z_{t-1}z'_{t-1} + b \cdot Q_{t-1}$; $R_t =$ standardized Q_t . Summary: ρ_t (mean off-diagonal correlation), disp_t (std off-diagonal). Rolling z-scores vs 60-day window. Five labels: broken ($\text{spread_stab}<30\%$), crisis ($\rho_z>2$), crowded ($\rho_z \in (1,2]$, $\text{disp}_z<0$), clustered ($\rho_z \in (1,2]$, $\text{disp}_z \geq 0$), normal (otherwise).

Book / Allocator Architecture

Book: owns α_{df} , cov_dict , optimizer params (γ , κ , λ , max_weight), vol params (target_vol , EWMA_HL). $\text{book.run}(\text{returns_df}) \rightarrow$ full backtest PnL, weights, Sharpe, max_dd , turnover, scaling stats. Books are independent — no shared state, no inter-book communication. Allocator: orchestrates Books. Current v1.0: sums PnL additively. Planned v2.0: apply regime-driven $\alpha_{\text{multipliers}}$ per book before calling $\text{book.run}()$. Why two books: reversal needs daily rebalancing + low κ ; momentum needs weekly + high κ . Mixing them in one optimizer creates turnover distortion for both signals.

Regime Mapping Logic

$\text{get_book_actions}(\text{regime_label}) \rightarrow \{\text{book_name}: \{\text{active}: \text{bool}, \text{alpha_multiplier}: \text{float}\}\}$. Normal: all books $1.0 \times$ alpha. Clustered: spread $1.3 \times$, others $0.8 \times - 1.0 \times$ (sector rotation $\rightarrow$ wider spreads). Crowded: spread $0.6 \times$ (all assets moving together $\rightarrow$ spreads compressed). Crisis: all $0.3 - 0.5 \times$. Broken: spread disabled (cointegration has failed). CRITICAL: multipliers applied PRE-optimizer (modify alpha input), not post-solve. Post-solve scaling is canceled by vol targeting re-inflation.

SECTION 5 — INTERVIEW Q&A; (35 QUESTIONS)

A. PM / High-Level

Q: What is the economic thesis of this strategy?

Commodity futures and sector ETFs share fundamental economic linkages — oil prices affect energy stocks, metals affect industrials — creating structural mean-reverting spreads. The thesis is that these deviations revert on a 1–4 week horizon and can be captured net of transaction costs. The cross-asset design allows genuine beta neutralization (unlike pure equity stat arb), leaving signals driven by sector-specific and commodity-specific factors.

Q: Why these specific asset classes? Why not pure equities?

ETFs carry equity beta and earnings risk; commodity futures carry roll yield and supply/demand risk. Combining them creates more genuine diversification for long-short stat arb than pure equities, because the residual (after beta neutralization) is driven by fundamentally different mechanisms. Also, the ETF-commodity pair relationships have economic backing — you can tell a story about why XLE and CL should be cointegrated — not just statistical coincidence.

Q: What is the capacity of this strategy?

At ~\$50–200M before market impact becomes material. Sector ETFs are liquid (easily \$1B+ ADV), but some commodity futures are thinner. At \$100M AUM, 15% target vol, 7% max per-asset position: peak notional per asset is \$7M — manageable for most futures. Scaling requires adding instruments or moving to more liquid markets (index futures, currency futures).

Q: How would you extend this to a larger universe?

Add international sector ETFs, currency futures (FX carry), bond futures (cross-asset macro dimension), more commodity contracts (metals, softs, grains). The Book/Allocator architecture scales naturally — add a new Book with its own alpha, covariance, and parameters. The Allocator handles combination. DCC scaling is $O(N^3)$, so a richer DCC specification or factor-based correlation model may be needed at $N > 50$.

B. Quant Researcher Questions

Q: Walk me through the cointegration signal construction.

For each (ETF, commodity) pair: rolling OLS (120d window) estimates hedge ratio. Three sequential gates: (1) Rolling ADF within window, $p < 0.10$ for stationarity. (2) ECM regression: $\Delta \text{spread} = \alpha + \rho \cdot \text{spread_lag}$; require $\rho \in (-1, 0)$. (3) Half-life $h = -\ln(2)/\ln(1+\rho) < 30$ days. If all pass: signal = rolling z-score of current spread deviation. If any gate fails: signal = 0. The three gates eliminate spurious cointegration from small-sample noise.

Q: Why rolling ADF instead of one-time cointegration test?

The original code ran ADF on the full sample — look-ahead bias of the worst kind. At date t , you'd be using data from t to T to decide whether to trade today. Cointegration also changes over time: oil-XLE was strongly cointegrated 2010–2018, broke during COVID, partially recovered. Rolling ADF adapts to the current regime and is strictly backward-looking.

Q: What does the Fama-MacBeth regression actually estimate?

At each rebalancing date t , it estimates how many basis points of forward return a unit of signal predicted, averaged over the past 52 weeks. FM beta $\approx IC \times \sigma_{\text{returns}}$. This converts dimensionless CS z-scores to

expected-return scale. Without FM alignment, the optimizer's risk term (in return² units) dominates the alpha term by ~50,000×, producing near-zero positions.

Q: Why clamp the FM beta at zero?

With 13 assets and ~7 usable cross-sections per week, the standard error on each cross-sectional OLS beta is large — roughly 50% of individual weekly estimates are negative by pure noise. Allowing negative FM betas would invert the signal direction (reversal becomes trend-following). Clamping at 0 is conservative: zero position that week rather than an inverted trade.

Q: Why does reversal fail in the combined optimizer?

Horizon mismatch. Reversal is a 5-day signal that decays in days; momentum is a 60-day signal that persists for weeks. The weekly optimizer with $\kappa=10$ creates high position inertia — correct for momentum (you want smooth, persistent positions) but wrong for reversal (you need rapid turnover to capture 5-day mean reversion). Combining them in one optimizer produces a κ calibrated incorrectly for both. Solution: separate Books with different rebalancing frequencies.

Q: Explain the optimizer objective and its parameters.

$\max \alpha'x - (\gamma/2)x'Sx - (\kappa/2)\|x - x_{\text{prev}}\|^2 - \lambda \cdot \|\text{sign}(x_{\text{prev}})\|$. $\gamma=20$: risk aversion (scales down positions relative to alpha). $\kappa=10$: inertia (penalizes deviations from previous positions). $\lambda=0.0002$: L1 cost (discourages exiting existing positions). Closed-form: $x = (\gamma\Sigma + \kappa I)^{-1}[\alpha + \kappa x_{\text{prev}} - \lambda \cdot \text{sign}(x_{\text{prev}})]$. Higher $\gamma \rightarrow$ smaller, lower-risk positions. Higher $\kappa \rightarrow$ slower, smoother position evolution.

Q: What's the role of the GJR leverage effect in GJR-GARCH?

Standard GARCH treats positive and negative shocks symmetrically. Empirically, negative returns increase future volatility more than equivalent positive returns — this is the leverage effect. GJR adds the term $\gamma \cdot I(r < 0) \cdot r^2$, which amplifies vol response to negative shocks. For commodities (supply disruptions, geopolitical shocks), the asymmetry is especially pronounced. The GJR standardized residuals are more appropriate inputs for DCC than standard GARCH residuals.

Q: How does the DCC model capture cross-asset correlation regime shifts?

DCC allows the $N \times N$ correlation matrix to evolve through time: $Q_t = (1-a-b)Q_{\infty} + a \cdot z_{t-1}z' + b \cdot Q_t$. In calm periods, Q_t reverts toward the unconditional mean Q_{∞} (parameter b drives persistence). During stress (high absolute residuals z), Q_t gets pulled toward extreme correlations. Standardizing $Q_t \rightarrow R_t$ gives the conditional correlation matrix. The mean off-diagonal element ρ_t captures whether assets are moving together — the core crisis indicator.

Q: Explain the vol targeting two-layer design.

Layer 1 (LW Σ): controls RELATIVE sizing — which assets to emphasize within the portfolio. Layer 2 (EWMA realized-vol): controls ABSOLUTE leverage — how much total risk to take. $\text{Scale} = \text{clip}((\text{target_vol}/RV)^2, 0.2, 1.5)$. $\text{Power}=2$ makes the response more aggressive. The separation is necessary because with $T/N \approx 9$, LW systematically underestimates portfolio vol by 30× if used directly for vol targeting. Using realized vol instead bypasses the model bias.

C. Architecture / Engineering

Q: Why a Book abstraction instead of a single optimizer?

Different strategies require different optimizer parameters, rebalancing frequencies, covariance windows, and vol targets. Reversal needs daily rebalancing ($\kappa=2$, 20d covariance window); momentum needs weekly rebalancing ($\kappa=10$, 120d window). One optimizer can't be optimally calibrated for both. Book encapsulates all

strategy-specific configuration — clean separation of concerns, independently testable, no shared state between strategies.

Q: How does the Allocator combine books?

Current v1.0: sums PnL additively (equivalent to sub-portfolios). The architectural design for v2.0: Allocator applies regime-driven α _multipliers before passing α _df to each Book. So in crisis regimes, spread book receives 40% alpha, optimizer naturally sizes down relative to normal. This is pre-optimizer conditioning, not post-solve scaling.

Q: Why is regime logic PRE-optimizer, not post-solve?

Post-solve scaling is canceled by the EWMA vol targeting. If you halve positions post-solve, realized vol drops, EWMA detects lower vol, vol targeting scales up by 2 $\times$, positions return to original size. The regime adjustment is exactly neutralized. Pre-optimizer alpha multipliers correctly modify the objective function: lower alpha $\rightarrow$ optimizer naturally chooses smaller positions while vol targeting responds to lower realized vol over time.

Q: What would you need to productionize this?

Five gaps: (1) Live data pipeline — currently flat CSV files; need real-time futures data with continuous contract adjustments. (2) Execution layer — $\lambda=0.0002$ is stylized; need VWAP/TWAP models and actual market impact estimates per instrument. (3) Walk-forward hyperparameter validation — γ , κ , EWMA halflife not validated out-of-sample. (4) Risk limits beyond MAX_WEIGHT — VaR/CVaR constraints, drawdown circuit breakers. (5) Monitoring — DCC convergence tracking, signal health metrics, book P&L; attribution in real time.

Q: Why keep run_baseline.py frozen instead of deleting it?

Regression baseline. After every architectural refactoring, run_engine.py should produce the same medium-term book PnL as run_baseline.py within floating-point tolerance. This prevents silent regressions — if the modular version's PnL diverges significantly from the monolithic version after a refactoring step, something broke in the extraction. The monolithic baseline is the ground truth for correctness validation.

D. Implementation / Criticism

Q: The DCC parameters are fit on the full sample — isn't that look-ahead bias?

Partially. The DCC parameters (a , b) are estimated on the full sample — minor look-ahead in the model parameters. However, the conditional correlation path R_t is computed recursively: R_t depends only on R_{t-1} and residuals up to t . So regime labels at time t don't use future covariance realizations. In production, parameters would be re-estimated rolling (quarterly). The parameter look-ahead is second-order — the directional exposure decisions are based on R_t , which is recursively filtered with no future information.

Q: Your optimizer uses post-solve projection for constraints — is that correct?

Technically sub-optimal. The closed-form solution is optimal for the unconstrained problem; post-solve projection onto the dollar-neutral and position-limit feasible set makes it sub-optimal for the constrained problem. When alphas are diversified and position limits are not too tight, the distortion is small. In production, a proper QP solver (scipy minimize, CVXPY) would handle constraints correctly. The current approach trades mathematical purity for computational simplicity and interpretability.

Q: Why power=2 in vol scaling?

Power=1 (scale = target_vol/RV) empirically failed to keep portfolio vol near target over extended elevated-vol periods. When $RV = 1.5 \times \text{target}$: power=1 gives scale=0.67, still leaving the portfolio at $\sim 1.3 \times$ target vol. Power=2 gives scale=0.44, much more aggressive deleveraging. Trade-off: power=2 with short EWMA halflife (2 weeks) can produce large weekly swings in leverage. The choice was validated empirically — max

drawdown dropped from -48.8% to -7.2% after this change.

Q: How robust is the cointegration structure across regimes?

Not robust to structural breaks. Oil-XLE cointegration was strong 2010–2018, broke during COVID (demand shock with sector-specific dynamics), partially recovered. The rolling ADF gate handles this automatically — when the spread is non-stationary in the rolling window, signal = 0. The 'broken' regime label ($< 30\%$ of pairs active) also captures systematic breakdown and disables the spread book. The gap is that recovery after a break is slow — the gate won't re-activate until the spread re-establishes stationarity for 120 days.

Q: What's the real IC of your signals?

FM alignment implicitly estimates weekly cross-sectional IC each rebalancing period. Typical values: momentum and spread ≈ 0.03 – 0.08 , reversal ≈ 0 (in the weekly book). Standalone Sharpe ratios: momentum ≈ 0.34 , spread ≈ 0.37 , reversal ≈ -0.38 . These are consistent with academic literature on weekly cross-sectional momentum in commodities (IC ≈ 0.04 – 0.10). The combined Sharpe of 0.254 reflects both signal quality and the transaction cost penalties from weekly rebalancing.

Q: Why does realized vol still exceed the 15% target at 27.8%?

Three compounding reasons. (1) LW with $T/N \approx 9.2$ underestimates portfolio vol — the model predicts smaller vol than the portfolio delivers, so the EWMA scaling is initially under-responsive. (2) EWMA half-life = 2 weeks creates lag: during vol spikes, 2 weeks of history doesn't fully capture the regime shift, leaving the portfolio over-sized for several weeks. (3) The underlying strategies are inherently volatile — FM-aligned alphas predict smaller expected returns than realized, causing the optimizer to size positions larger than optimal.

E. Quant Finance / Risk Questions

Q: What is the fundamental risk of a stat arb strategy like this?

Model risk is primary: cointegration relationships are estimated historically and may break out-of-sample. Stat arb strategies are short convergence risk — they provide liquidity into drawdowns. When spreads diverge (structural shift, crisis), the strategy is exactly positioned to be hurt: long the cheap leg, short the rich leg of a spread that keeps widening. The ECM half-life filter and 'broken' regime label partially mitigate this, but cannot prevent temporary large drawdowns during rapid regime shifts.

Q: Explain the risk decomposition of the portfolio.

Long idiosyncratic risk (beta-neutral $\rightarrow$ market risk removed). Short correlation risk (spread profits when cross-asset correlations are low; loses when they spike toward 1). Long liquidity risk (reversal book provides liquidity in the short term). Short regime-shift risk (positions built on historical cointegration can be wrong if the regime changes). DCC-GARCH partially hedges the correlation risk by reducing exposure in high-correlation regimes before the drawdown occurs.

Q: How would you think about drawdown risk?

Phase 2c max drawdown: -7.2% in-sample over 9.5 years. Out-of-sample, expect 2–3 $\times$ larger. The EWMA vol targeting limits drawdown growth by deleveraging as vol rises, but with a 2-week lag. During fast-moving events (March 2020), the portfolio can lose more than target vol implies before the EWMA catches up. Circuit breakers (e.g., suspend trading if 30-day drawdown $> 10\%$) would be needed in production as a hard floor.

Q: Why didn't you include carry?

Carry for futures means roll-yield (front month vs second month settlement price difference). Without term-structure data (CME settlement prices), there is no correct carry proxy. The original code used 30-day rolling residual returns as a carry proxy — this captures short-term trend, not carry. Rather than implement a

flawed signal, the decision was to remove carry entirely. A correct implementation would require an additional data feed.

SECTION 6 — HONEST WEAKNESSES

Be prepared to discuss these candidly. Intellectual honesty about limitations is a signal of maturity. Frame each weakness with (1) what it is, (2) why it matters, (3) what would fix it.

1. Cointegration Instability

Rolling ADF with 120-day window cannot distinguish 'stressed but intact' from 'permanently broken' cointegration. During COVID, energy-commodity-ETF relationships broke for months. The gate correctly zeros out the signal, but re-entry after recovery is slow (needs 120 days of re-established stationarity). Fix: adaptive window length, or a Bayesian regime model for pair stability.

2. Small Cross-Section (N=13)

Fama-MacBeth cross-sectional OLS with ~7 usable assets per week has enormous standard errors. The FM beta floor is a blunt patch. With N=13, diversification is limited and signal estimation is noisy. Fix: expand universe to 30–50 instruments (international ETFs, more commodity contracts).

3. Covariance Instability

LW with $T/N \approx 9.2$ applies aggressive shrinkage toward identity. The model systematically underestimates correlations during stress — exactly when risk management matters most. Model-implied vol $\approx$ realized vol only on average; at tail events, the model is off by 2–5 $\times$. Fix: factor-based covariance model (BARRA-style) or longer lookback.

4. Transaction Cost Realism

$\lambda = 0.0002$ is stylized. At \$100M AUM with 30-40% weekly one-way turnover, actual slippage (2–10bp per trade in commodity futures) could be \$2–6M/year — material relative to expected alpha. Fix: calibrate λ to actual spread $\times$ ADV for each instrument; implement market-impact model.

5. Execution Assumptions

All 13 assets are treated as equally liquid. Natural gas futures have $\approx 30\%$ of crude oil ADV; agricultural contracts have $\approx 20\%$. In practice, position limits should scale with ADV. Fix: ADV-adjusted max position size per instrument.

6. Regime Mapping Not Validated

The regime $\rightarrow$ book action mapping (e.g., spread 1.3 $\times$ in clustered regime) encodes economic hypotheses that have NOT been empirically validated. These multipliers could easily be wrong. Fix: walk-forward validation of regime-conditional performance; empirically estimate optimal multipliers.

7. Vol Targeting Lag

EWMA half-life=2 weeks creates a systematic lag response. During fast vol spikes, the portfolio runs 1.5–2 $\times$ target vol for 2–4 weeks. Power=2 partially compensates but also creates leverage instability. Fix: shorter EWMA half-life (1 week) with dampening; or VIX-level overlay as a leading indicator.

8. Residual Look-Ahead (DCC Parameters)

DCC parameters (a, b) are fit once on the full sample — minor look-ahead in parameter values. R_t computation is recursive (no future data in conditional path), so regime labels at time t are look-ahead-free in the signal. But DCC parameters calibrated to include tail events like 2020 may overstate the reaction speed in out-of-sample production. Fix: rolling quarterly DCC refitting.

9. Optimizer Constraints via Projection

Post-solve projection (dollar neutrality + position clipping) is not true constrained optimization. When alphas are concentrated, projection can significantly distort the optimizer's solution. Fix: replace closed-form + projection with

a proper QP (CVXPY or `scipy.optimize.minimize`).

10. Futures Roll Assumptions

Continuous front-month contracts introduce roll returns (positive in backwardation, negative in contango) that are not separated from price returns. The strategy doesn't explicitly model roll timing, which can introduce artifacts especially for seasonal commodities (natural gas in winter). Fix: use Panama or ratio-adjusted continuous contracts with explicit roll cost accounting.

SECTION 7 — MUST-KNOW CONCEPTS

Concept	30-Second Explanation
Fama-MacBeth	Cross-sectional OLS each period; average β over time. FM beta $\approx$ IC $\times$ σ_{returns} . Converts z-scores to $E[r]$ units.
ADF Test	H_0 : unit root. Reject $\rightarrow$ stationary. Rolling = backward-looking only. $p < 0.10$ = pass.
ECM half-life	$\Delta \text{spread} = \alpha + \rho \cdot \text{lag} + \varepsilon$. $\rho \in (-1, 0) \rightarrow$ mean-reverting. $h = -\ln 2 / \ln(1 + \rho)$. Filter: $h < 30d$.
Ledoit-Wolf	$\Sigma = (1 - \alpha)S + \alpha(\mu/N)I$. Optimal shrinkage analytically. Small $T/N \rightarrow$ aggressive shrinkage. Controls relative sizing.
GJR-GARCH	GARCH + asymmetric $\gamma I(r < 0)r^2$ term. Negative shocks inflate vol more. Leverage effect.
DCC	$Q_t = (1 - a - b)Q + a \cdot zz' + b \cdot Q_t$; normalize $\rightarrow R_t$. Time-varying $N \times N$ correlations. Captures crisis spikes.
Dollar neutrality	$\sum x_i = 0$. Long = Short. Removes market drift. Applied post-solve: $x := x - \text{mean}(x)$.
Beta neutralization	$r_{\text{asset}} = \alpha + \beta \cdot r_{\text{mkt}} + \varepsilon$. Use ε for signals. Removes common market factor.
κ (inertia)	Penalizes $\ x - x_{\text{prev}}\ ^2$. Higher $\kappa \rightarrow$ lower turnover, slower position adjustment.
Vol targeting	Scale = clip((target_vol/RV)^power, min, max). Two layers: LW (relative), EWMA (absolute).
FM beta floor = 0	Clamp negative β to 0 (estimation noise in small N). Zero position vs. inverted signal.
Book vs Allocator	Book = one strategy's full pipeline (owns alpha, Σ , optimizer, vol target). Allocator = orchestrates Books.
Regime labels	5 states: normal/clustered/crowded/crisis/broken. Crisis=cut all. Broken=spread disabled. Pre-optimizer.
Why two books?	Reversal (5d signal, daily rebalancing) + momentum (60d signal, weekly) can't share one κ .
Log vs simple returns	Log: aggregate over time. Simple: aggregate over assets. Commodity moves: wrong choice flips PnL sign.

Must Be Ready to Draw / Derive

- Signal pipeline flowchart: Data $\rightarrow$ Beta Neutral $\rightarrow$ [Rev | Mom | Spread] $\rightarrow$ FM Align $\rightarrow$ Sharpe Combine $\rightarrow$ Optimizer $\rightarrow$ Vol Scale $\rightarrow$ Weights
- Optimizer objective: write full equation, identify each term, state closed-form solution.
- ECM filter: Δspread vs. $\text{spread}_{\text{lag}}$ scatter, slope = ρ , show $h = -\ln 2 / \ln(1 + \rho)$.
- DCC diagram: Returns $\rightarrow$ GJR-GARCH (per asset) $\rightarrow Z \rightarrow DCC \rightarrow R_t \rightarrow \rho_t \rightarrow$ regime label.
- Vol targeting two-layer diagram: LW Σ (relative) $\rightarrow$ EWMA scale (absolute) $\rightarrow$ MAX_WEIGHT clip.

Must Understand the Evolution

- Carry was wrong $\rightarrow$ removed (not fixed).
- Full-sample ADF $\rightarrow$ rolling ADF (look-ahead bias fix).
- Static risk-parity $\rightarrow$ rolling Sharpe combination (another look-ahead fix).
- FM alignment solves the 14,000 $\times$ signal-scale mismatch.
- LW on raw returns inflated Σ 30 $\times$ $\rightarrow$ broke vol targeting $\rightarrow$ fix: use residual returns.

- LW on residual returns still underestimates vol ($T/N \approx 9$) → separate vol control layers.
- Monolithic → Book/Allocator (modularity, testability, multi-strategy support).
- Reversal excluded from combined alpha (horizon mismatch: 5d signal $\neq$ weekly optimizer).
- Post-optimizer regime scaling is wrong (vol targeting cancels it) → pre-optimizer alpha multipliers.
- Log returns break short-term commodity PnL → use simple returns in short-term book.